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Glider model

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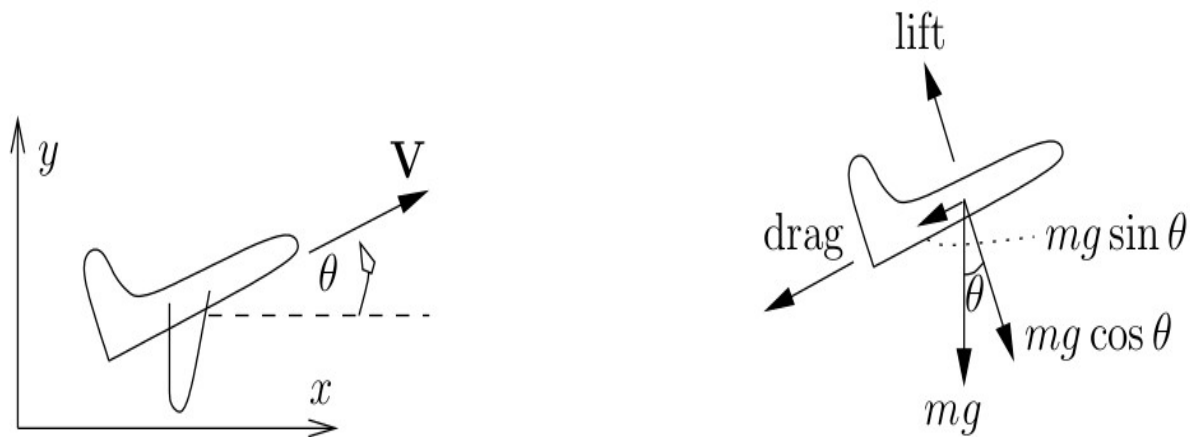
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The trajectory of a glider

A simple model to describe the flight of a glider.

Glider model

The glider flies under an angle θ with the horizon and with (scalar) velocity V .



$$m \frac{dV}{dt} = -c_D V^2 - mg \sin(\theta)$$

$$mV \frac{d\theta}{dt} = +c_L V^2 - mg \cos(\theta)$$

Tasks for the project in Module 4

Finish the modelling for the full basic model. First a repeat of the optional tasks in module 3; if you have not done them there, do them here.

1. Choose one or more sets of initial conditions $V(0)$ and $\theta(0)$. Do not choose the equilibrium values for both of the initial conditions $V(0)$ and $\theta(0)$, because nothing will happen then! Implement Euler's method for the system, where both V and θ change in time. If you have decided on different stepsizes for the calculations of the separate $V(t)$ and $\theta(t)$, for the system you need to calculate with the minimum of the two stepsizes. If in doubt check the errors again.
2. When you can simulate the solutions of the system for $V(t)$ and $\theta(t)$, you also want to see what the trajectory of the glider is. The trajectory through the sky is described by the horizontal position $x(t)$ and the vertical position, or height $y(t)$. x and y are described by first-order differential equations of the following form:

$$\frac{dx}{dt} = f_x(V, \theta)$$

$$\frac{dy}{dt} = f_y(V, \theta).$$

Derive expressions for $f_x(V, \theta)$ and $f_y(V, \theta)$ yourself.

3. Include the two differential equations for x and y into the system in your simulation program (so you solve 4 dependent variables at the same time). Start plotting trajectories. Note that different scales in the figure for the x and y make it difficult to interpret the angles in the trajectory.

4. Calculate the position $(V_{\text{eq}}, \theta_{\text{eq}})$ of the equilibrium point analytically. Use a simple relation between a sine and a cosine to find an explicit expression for V_{eq} . In the expression for the θ_{eq} you may approximate $\tan(\theta) \approx \theta$, because it is such a small angle.
5. Find a reasonable values for c_D and c_L . There are different ways for finding a value:
 - Compare the expressions for $(V_{\text{eq}}, \theta_{\text{eq}})$ with the values for your glider.
 - Numerical simulation experiments and see whether the resulting trajectories are plausible.
 - Maybe you can find information on c_D or c_L elsewhere. Take care though that you only use constants that apply to the same modelling term. For example, check that the units agree.
 - Calculation of the eigenvalues of the Jacobian matrix in the equilibrium point. If the solution oscillates, determine the period of the oscillations and set c_D or c_L to a value such that the period of the oscillations is plausible. In that calculation, you can use that for small angles $\cos(\theta) \approx 1$, and $\sin(\theta) \approx \theta$.

Ideally, you use more methods to determine the constants. On the other hand, maybe you want to press on, and focus on other parts of the modelling process. Then you use just one method.

New points for module 4:

6. Simulate some flights with different initial conditions. Generate graphs of $V(t)$ and $\theta(t)$ versus time, and graphs of trajectories y versus x . Are these values reasonable for your plane?
7. Do you see oscillations? Is the period of the oscillations comparable to the period you predicted analytically in task 5?
8. Generate graphs of the solutions in the phase plane: θ versus V . Use different sets of initial conditions, and plot all solutions in the same phase plane.
9. Make loopings with your virtual plane (higher initial velocity, higher angle, maybe a lower drag). Plot the xy —trajectories with loopings. Plot simulations with loopings also in the $V\theta$ -phase plane.

Extending your model

After you have written and submitted your first report, in module 5 you have to go beyond the basic model. You should start thinking about what you would like to do then. What specific problem about the glider do you want to solve? That can be a pressing real-life problem, or you could investigate something you made up. What if ...?

So you could investigate what happens when there is some intervention: extra wind, a thermal, the use of a height rudder, braking by extending air brakes. An intervention can be at $t = 0$, or you could simulate it to take effect after a certain time by including an if statement: If $t \geq 50$: and then your intervention for example changes the derivatives. Then your final modelling problem could be What is the influence of ...?

You could also try to simulate different strategies for contests: the longest flight, the fastest flight, or try to simulate landing smoothly on an airstrip.

If you know a little more about aerodynamics, you may be dissatisfied with this model for your glider, and you may alter one or more of the terms, for example make the constants depend on the angle or velocity. And then you have to show that that model is better than the simple one. So then your research problem is to find a better model for this glider.

This extension of the model is for module 5. Finish your work on the basic model first and write and submit the first report before June 27th.

After the page for the medicine problem, you will find a page for questions you might have about the project.

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